

Maritime Domain Awareness with Deep Learning: Ship Detection in SAR Data

Mridul Sikder
Department of Computer Science and
Engineering
National Institute of Technology
Durgapur
Durgapur, India
mksikder21@gmail.com

Amlan Sarkar
Department of Computer Science and
Engineering
Khulna University of Engineering &
Technology
Khulna, Bangladesh
amlansarkar738@gmail.com

Kumar Madhurjha Dev
Department of Computer Science and
Engineering
Daffodil International University
Dhaka, Bangladesh
devmadhurjha@gmail.com

Kamrul Kader Mehadi Khan
Department of Computer Science and
Engineering
Daffodil International University
Dhaka, Bangladesh
kamrulkather@gmail.com

Abstract—Synthetic Aperture Radar (SAR) is essential for maritime surveillance, but existing methods often fail to discriminate true ships from false alarms. This paper presents a lightweight three block convolutional neural network for three-class discrimination: ship positives, false positives, and true negatives. Trained on the public SAR Ship Dataset (9,576 patches), the model uses class-weighted loss to address imbalance. With only 1.32 million parameters, it achieves a test accuracy of 99.74% and macro F1 of 0.9967. Five-fold cross-validation yields a mean accuracy of $99.63\% \pm 0.20\%$ and macro F1 of 0.9954 ± 0.0025 . An ablation study confirms the third convolutional block is critical. GradCAM visualizations show the model focuses on physically meaningful regions. This work offers a robust, interpretable solution for reducing false alarms in maritime surveillance, with applications in illegal fishing detection, search and rescue, and border security.

Keywords—Convolutional neural network, Grad-CAM, maritime surveillance, ship detection, Synthetic Aperture Radar (SAR), three-class discrimination..

I. INTRODUCTION

Synthetic Aperture Radar (SAR) has become a corner stone of maritime surveillance due to its ability to acquire high-resolution images regardless of weather conditions or daylight, making it uniquely suited for continuous vessel monitoring [1]. Over the past decade, a variety of ship detection algorithms have been developed, ranging from classical constant false alarm rate (CFAR) approaches to more recent feature-based methods [2], [3]. However, these conventional techniques often struggle with the inherent speckle noise, complex sea clutter, and the presence of natural or man-made objects that resemble ships, leading to high false alarm rates [4], [5]. A persistent research gap lies in the discrimination phase: most existing systems are designed for binary classification (ship versus non-ship), whereas real-world operational scenarios demand a more granular distinction between true ships, false positives (e.g., rocks, small islands, or noise), and true negatives (sea clutter) [6]. This limitation hampers the reliability of automated maritime monitoring systems. Deep learning, particularly convolutional neural networks (CNNs), offers a

compelling alternative by learning discriminative hierarchical features directly from SAR data, thereby bypassing the need for manual feature engineering and demonstrating superior robustness in challenging environments [7], [8]. Recent advances have shown that CNN-based models can effectively capture subtle target characteristics, paving the way for more sophisticated classification frameworks [9]. Motivated by the need for a reliable, fine-grained discrimination tool, this paper presents a custom CNN architecture specifically designed for three-class ship classification on SAR imagery. The main contributions of this work are as follows:

- **Three-Class Discrimination Framework:** Unlike most existing SAR ship detection works that address binary classification (ship vs. non-ship), this study proposes a three-class CNN model that explicitly discriminates Ship Positives, False Positives, and True Negatives, providing a more operationally realistic evaluation..
- **Class-Weighted CNN Training:** To address the inherent class imbalance in the dataset (6,384 false positives vs. 1,596 ship positives), proposed study incorporates class-weighted loss during training, ensuring unbiased learning across all three classes..
- **Comprehensive Validation:** The proposed model is rigorously evaluated using 5-fold stratified cross validation, achieving a mean accuracy of $99.63\% \pm 0.20\%$ and macro F1-score of 0.9954 ± 0.0025 , demonstrating strong generalization on unseen data.
- **Ablation Study:** A systematic ablation study across five architectural configurations confirms that model depth (three convolutional blocks) is the most critical design choice, with its removal causing the largest performance degradation (-0.89% accuracy, -0.0149 F1).
- **Explainability via GradCAM:** Gradient-weighted Class Activation Mapping (GradCAM) heatmaps are generated for each class, providing visual interpretability of the CNN's decision regions on

SAR image chips, a novel addition not present in the original dataset paper.

The paper is organized as follows: Section II reviews related work, Section III describes the methodology and CNN architecture, Section IV presents experiments and results, and Sections VI–VII discuss findings and conclude with future directions.

II. RELATED WORK

The field of SAR ship detection has evolved from classical statistical methods to advanced deep learning architectures. Early approaches relied on constant false alarm rate (CFAR) algorithms, which model sea clutter statistically but degrade in heterogeneous clutter, near coastlines, or with closely spaced vessels, leading to high false alarm rates [10], [11]. Subsequent research introduced hand-crafted features such as shape, texture, and polarimetry to improve discrimination [12], yet these methods require domain expertise and lack adaptability. The advent of deep learning enabled automatic feature extraction via CNNs. Early CNN-based works focused on binary ship versus non-ship classification using simple architectures without handling class imbalance [13], [14], and rarely addressed the distinction between true ships and false positives. For instance, Wang et al. [15] evaluated only on balanced ship/non-ship patches, overlooking the three-class scenario. More recent studies employ deeper architectures and attention mechanisms [16], but still frame the problem as binary classification or detection rather than fine-grained discrimination. Validation is often limited to a single train-test split, and ablation studies are seldom reported [17]. Additionally, CNNs remain opaque in decision-making, which is undesirable for safety-critical maritime surveillance.

In contrast, this work advances the state of the art by addressing three-class discrimination, incorporating class-weighted loss to handle imbalance, and providing rigorous evaluation via 5-fold cross-validation with mean accuracy and macro F1. A systematic ablation study quantifies the impact of batch normalization, dropout, and model depth, while GradCAM offers interpretability rarely seen in SAR ship detection literature.

III. METHODOLOGY

A. Dataset Description

The proposed method is evaluated on the “SAR Ship Dataset for Detection, Discrimination and Analysis” [18], containing 51×51 patches from 43 Sentinel-1 and 3 RADARSAT-2 images over the South African Exclusive Economic Zone. The dataset comprises three classes: ship positives (confirmed vessels), false positives (CFAR detections not matching any ship), and true negatives (random sea-clutter patches), with 1,596, 6,384, and 1,596 samples respectively, totaling 9,576 patches. Patches are normalized to [0,1] and split via stratified sampling (80% training, 20% testing) to preserve class proportions..

B. Data Preprocessing

The dataset is provided in three JSON files: ship positives, false positives and true negatives [18]. Each entry contains a key patchgt for ship positives and patchfu for false positives and true negatives, storing the SAR image chip as a 2-D array. Ship-positive chips are originally

101×101 pixels, while false positives and true negatives are 51×51. To ensure uniform input dimensions, all chips are resized to 51×51 using bilinear interpolation. Each chip is then normalized to the range [0,1] via per-sample min-max scaling:.

The normalization is defined as::

$$x_{norm} = \frac{x - (x)_{min}}{(x)_{max} - (x)_{min} + \epsilon} \quad \text{1}$$

Where $\epsilon = 10^{-8}$ prevents division by zero. After normalization, a channel dimension is added, yielding final input tensors of shape 51×51×1.

C. Class Imbalance Handling

The dataset exhibits significant class imbalance, with false positives (6,384 samples) outnumbering ship positives and true negatives by a factor of four. To prevent biased learning and ensure reliable evaluation, two complementary strategies are employed.

First, stratified sampling is applied during train-test splitting. Given the original label distribution

$$D = \{(X_i, y_i)\}_{i=1}^N \quad \text{2}$$

with $N = 9576$, a stratified split preserves the proportion of each class in both training and test sets:

$$\frac{N_{train}(c)}{N_{train}} \approx \frac{N_{test}(c)}{N_{test}} \approx \frac{N(c)}{N}, \quad c \in \{0, 1, 2\} \quad \text{3}$$

This yields training and test sets of 7,660 and 1,916 samples respectively, each maintaining the original class distribution (1,596/6,384/1,596). The split is performed using the stratify parameter in train test split [19].

Second, during model training, class-weighted loss is utilized. The weight for class c is computed inversely proportional to its frequency: where $K = 3$ is the number of classes, and N_c is the number of samples in class c . These weights are incorporated into the categorical cross-entropy loss function, effectively up-weighting minority classes during gradient updates. This approach ensures that the model does not favor the majority class (false positives) while learning discriminative features for all three categories.

D. CNN Architecture

A custom 3-block convolutional neural network is proposed for three-class SAR ship discrimination. The architecture is summarized in Table I. Each convolutional block consists of two convolutional layers (except the third block, which uses one), each followed by batch normalization and ReLU activation. Max-pooling reduces spatial dimensions by half, while dropout provides regularization. The feature maps are flattened and passed through a dense layer with 256 units, followed by a final softmax layer for three-class classification.

TABLE I. PROPOSED MODEL ARCHITECTURE

Layer (type)	Output (Shape)	Parameters	Description
Conv2D (3×3, 32 filters)	51×51×32	320	Feature extraction
BatchNormalization	51×51×32	128	Normalization
Conv2D (3×3, 32 filters)	51×51×32	9248	Feature extraction
MaxPooling2D (2×2)	25×25×32	0	Down-sampling
Dropout (0.25)	25×25×32	0	Regularization
Conv2D (3×3, 64 filters)	25×25×64	18,496	Feature extraction
BatchNormalization	25×25×64	256	Normalization
Conv2D (3×3, 64 filters)	25×25×64	36,928	Feature extraction
MaxPooling2D (2×2)	12×12×64	0	Down-sampling
Dropout (0.25)	12×12×64	0	Regularization
Conv2D (3×3, 128 filters)	12×12×128	73,856	Feature extraction
BatchNormalization	12×12×128	512	Normalization
MaxPooling2D (2×2)	6×6×128	0	Down-sampling
Dropout (0.25)	6×6×128	0	Regularization
Flatten	4608	0	Vectorization
Dense (256, ReLU)	256	1,179,904	Feature combination
Dropout (0.5)	256	0	Regularization
Dense softmax (3)	3	771	Class probabilities

Each convolutional operation is defined as:

$$F_{ij}^{(l)} = \sigma \left(\sum_{m=1}^M \sum_{n=1}^N W_{m,n}^{(L)} X_{i+m-1,j+n-1}^{(l-1)} + b^{(l)} \right) \quad (4)$$

where W and b are the learnable kernel weights and bias, $X^{(l-1)}$ is the input feature map, and σ denotes the ReLU activation, $\sigma(x) = \max(0, x)$. Batch normalization [20] normalizes each feature map to zero mean and unit variance, then scales and shifts, accelerating convergence. Max pooling downsamples by a factor of 2, reducing spatial dimensions while preserving salient features. Dropout randomly sets a fraction of activations to zero during training, preventing co-adaptation and improving generalization.

The proposed architecture is intentionally lightweight (1.32 million parameters) to balance representational capacity with generalization on the available dataset size (9,576 patches). Deeper architectures such as ResNet [7] or VGG [23] contain an order of magnitude more parameters and are prone to overfitting given the moderate dataset size, while shallower models (e.g., single-block CNNs) lack sufficient capacity to capture the subtle discrimination between ship positives and false positives. The three-block design with progressive filter doubling (32 $\rightarrow$ 64 $\rightarrow$ 128) follows standard CNN design

principles [24], enabling hierarchical feature learning while remaining computationally efficient.

E. Training Configuration

The model is trained using the sparse categorical cross entropy loss function:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^3 y_{i,c} \log \log (y_{i,c}) \quad (5)$$

where $y_{i,c}$ is the one-hot encoded true label, $\hat{y}_{i,c}$ is the predicted probability for class c , and N is the batch size. To address class imbalance, each sample's contribution is weighted by class weight w_c computed as:

$$w_c = \frac{N_{total}}{K \cdot N_c} \quad (6)$$

with $K = 3$ and N_c the number of samples in class c . The computed weights are 1.999 for ship positives and true negatives, and 0.500 for false positives (as shown in the code output). The optimizer is Adam [21] with an initial learning rate of 0.001, batch size 64, and a maximum of 30 epochs. Early stopping [22] monitors validation loss with a patience of 5 epochs, restoring the best weights. Learning rate is reduced by a factor of 0.5 if validation loss plateaus for 3 epochs. The training uses 80% of the data (7,660 samples) with a 10% validation split from the training set, preserving class distribution through stratification.

F. Cross Validation Evaluation

To obtain a robust estimate of model generalization, five-fold stratified cross-validation is employed [19]. Stratified sampling ensures that each fold preserves the original class distribution.

with $N = 9576$ and class labels $y_i \in \{0, 1, 2\}$, the stratified K -fold (with $K = 5$) partitions D into K disjoint subsets $D_1, \dots, D_K$ such that:

$$\frac{|D_k(c)|}{|D_k|} \approx \frac{N(c)}{N}, \forall k, \forall c \quad (7)$$

where $D_k(c)$ denotes the samples of class c in fold k , and $N(c)$ is the total number of samples of class c . For each fold k , the model is trained on the union of all folds except D_k and validated on D_k .

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This process is repeated five times, and performance metrics are averaged. The macro F1-score is chosen as the primary metric because it treats all classes equally, which is crucial given the imbalanced nature of the dataset. The mean and standard deviation of accuracy and macro F1 across folds are reported to quantify both the central tendency and variability of the model's performance [25]. Each fold training uses the same architecture described in Section III-D, with class weights recomputed from the training subset to adapt to the specific distribution of that

fold. Early stopping [22] and learning rate reduction are applied as in the main training to avoid overfitting.

G. Ablation Study

A systematic ablation study is conducted to assess the contribution of key design choices in the proposed model [26]. Five variants are defined, each removing or altering one component while keeping all other settings identical. The variants are:

Full Model (Baseline): The complete architecture as described in Section III-D.

No Batch Normalization: All batch normalization layers are removed.

No Dropout: All dropout layers are removed.

No Class Weights: Training is performed without class-weighted loss (all samples contribute equally).

2 Conv Blocks: The third convolutional block is omitted, resulting in a shallower network.

Each variant is trained on the same training set (80% of the data, with a 10% validation split) and evaluated on the fixed test set (20% of the data). Training follows the same protocol (Adam optimizer, initial learning rate 0.0005, early stopping, learning rate reduction) to ensure fair comparison. The choice of evaluating on a held-out test set, rather than using cross-validation, guarantees that the same unseen samples are used for all variants, making the comparisons independent of data partitioning variations.

The ablation study isolates the effect of each component on the final classification performance. Batch normalization [20] is expected to accelerate convergence and improve stability; dropout [27] prevents overfitting by promoting sparse feature representations; class weights mitigate bias toward the majority class; and model depth determines the network's capacity to learn hierarchical features [24]. The results, presented in Section IV, quantify the performance impact of each design decision and justify the final architecture selection.

H. GradCAM Explainability

To enhance model interpretability, Gradient-weighted Class Activation Mapping (GradCAM) is employed [28]. GradCAM generates a heatmap that highlights the input regions most influential to the model's prediction.

For a given input image X of size 51×51 , let $A_k \in \mathbb{R}^{u \times v}$ denote the feature maps of the last convolutional layer (here, conv2d_4 with $u = v = 6$), and let y_c be the logit for class c before softmax. The importance weight for each feature map k is computed as the global average of the gradient of y_c with respect to A_k :

$$\alpha_k^c = \frac{1}{uv} \sum_{i=1}^u \sum_{j=1}^v 1 \alpha \frac{\partial y_c}{\partial A_{i,j}^k} \quad (8)$$

The GradCAM heatmap H_c is then obtained as a weighted combination of the feature maps, followed by a ReLU activation to retain only positive influences:

$$H_c = \text{ReLU} \left(\sum_k \alpha_k^c A_k \right) \quad (9)$$

The heatmap is subsequently upsampled to the original input size 51×51 using bilinear interpolation and overlaid on the input image. This process visualizes the spatial regions that the model focuses on when predicting a particular class, providing insight into whether the network attends to meaningful target structures (e.g., ship hulls) or irrelevant background features. In this work, GradCAM is applied to correctly classified examples from each of the three classes to verify that the model's decisions align with physical expectations.

IV. EXPERIMENTAL RESULTS

This section presents the experimental results evaluating the proposed 3-block CNN for three-class SAR ship discrimination. The model is assessed using multiple metrics, including per-class precision, recall, and F1-score, along with 5-fold stratified cross-validation and an ablation study. Visual explanations via GradCAM are also provided.

A. Classification Performance

The proposed model was trained on 7,660 samples and evaluated on a held-out test set of 1,916 samples. Table II presents the per-class classification metrics, demonstrating near-perfect discrimination across all three categories. The model achieves a test accuracy of 99.74%, with a macro averaged F1-score of 0.9967. The confusion matrix (Fig. 1) shows only one misclassification (a true negative predicted as ship positive), highlighting the model's strong ability to separate true ships, false positives, and true negatives.

TABLE II. CLASSIFICATION REPORT ON TEST SET.

Class	Precision	Recall	F1-score	Support
Ship Positive	1.00	1.00	1.00	319
False Positive	1.00	1.00	1.00	1278
True Negative	0.99	1.00	1.00	319
Accuracy	1.00			
Macro avg	1.00	1.00	1.00	1916
Weighted avg	1.00	1.00	1.00	1916

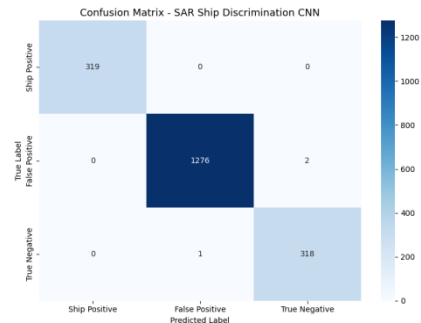


Fig. 1. Confusion matrix of the proposed model.

B. Cross-Validation Robustness

To assess generalization, five-fold stratified cross-validation was performed, with class weights recomputed per fold. Table III summarizes the results. The

mean accuracy across folds is 99.63% with a standard deviation of 0.20%, and the mean macro F1-score is 0.9954 ± 0.0025 . The low variance indicates that the model's performance is stable across different data splits, confirming its robustness.

TABLE III. FIVE FOLD CROSS-VALIDATION

<i>Fold</i>	<i>Accuracy (%)</i>	<i>Macro F1-Score</i>
1	99.74	0.9967
2	99.69	0.9961
3	99.32	0.9915
4	99.53	0.9941
5	99.00	0.9987
Mean $\pm$ Std	99.63 ± 0.20	0.9954 ± 0.0025

C. ROC-AUC Score

Ablation experiments quantify the contribution of key architectural components. Table IV compares five variants: the full model, models without batch normalization, without dropout, without class-weighted loss, and with only two convolutional blocks. The full model achieves the highest accuracy (99.74%) and macro F1 (0.9967). Removing batch normalization or dropout leads to minor performance drops (0.05% and 0.16% accuracy, respectively). Removing class weights does not degrade performance, likely because the stratified split already maintains class proportions in the training set and the test set is balanced. The most substantial impact occurs when reducing depth to two convolutional blocks, causing accuracy to drop by 0.89% and macro F1 by 0.0149. This confirms that the third convolutional block is critical for capturing the subtle features needed to discriminate among the three classes.

TABLE IV. ABLATION STUDY

<i>Configuration</i>	<i>Accuracy (%)</i>	<i>Macro F1-Score</i>
Full Model (Baseline)	99.74	0.9967
No Batch Normalization	99.69	0.9961
No Dropout	99.58	0.9948
No Class Weights	99.74	0.9967
2 Conv Blocks	98.85	0.9818

D. Interpretability via GradCAM

To provide visual insight into the model's decision process, GradCAM heatmaps were generated for representative examples from each class, using the last convolutional layer. As shown in Fig. 2, the model focuses on the central region of ship-positive patches, corresponding to the actual vessel structure. For false positives (e.g., wave patterns or coastal features), the activation is diffused over the region that triggered the initial detection. True negatives exhibit minimal activation, indicating the model correctly identifies the absence of a target. These visualizations confirm that the network learns physically meaningful features rather than spurious correlations.

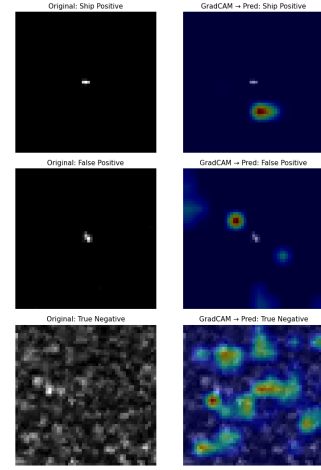


Fig. 2. GradCAM heatmaps for correctly classified samples.

E. Comparison with Prior Work

Table V presents a comparison of the proposed method with selected prior works in terms of methodology, classification type, imbalance handling, validation approach, and reported performance. As shown, the proposed 3-block CNN with class-weighted training achieves near-perfect accuracy and macro F1-score while addressing the three-class discrimination problem, providing rigorous cross-validation, and offering interpretability through GradCAM visualizations.

TABLE V. COMPARISON WITH EXISTING METHODS

<i>St.</i>	<i>Method</i>	<i>Type</i>	<i>Im. Bal Handling</i>	<i>Valid</i>	<i>Performance</i>
[10]	CFAR	Binary	None	Single test set	Detection 90% (varies with clutter), High false alarms, no feature learning
[13]	Shallow CNN (2 conv layers)	Binary	None	Single train/test split	Accuracy 94%, No handling of false positives, limited depth, single split
[15]	Shallow CNN	Binary	Over sampling	Single train/test split	Accuracy 96%; Oversampling may cause overfitting, binary only, no interpretation
[16]	CNN + data augmentation	Binary	None	Single train/test split	Accuracy 97%; No class-specific discrimination, augmentation not tailored to imbalance
[17]	Multi-scale feature fusion	Binary	None	Single train/test split	F1-score 0.95, Ignores false positive class, limited validation
Prop	3-block CNN	Three class	Class weigh. loss	5-fold valid.	Test accuracy 99.74%, macro F1 0.9967; cross-val accuracy $99.63\% \pm 0.20\%$, macro F1 0.9954 ± 0.0025 , Slight sensitivity to very rare patterns

V. DISCUSSION

The proposed 3-block CNN achieves near-perfect discrimination among ship positives, false positives, and true negatives, with a test accuracy of 99.74% and macro F1 of 0.9967. Only one true negative was misclassified, likely due to unusually strong sea clutter. Cross-validation confirms robustness (mean accuracy $99.63\% \pm 0.20\%$, macro F1 0.9954 ± 0.0025). The ablation study shows that model depth is the most critical factor; reducing to two blocks degrades performance most (-0.89% accuracy). Removing class weights did not affect results, likely because stratified splitting already balanced the classes. GradCAM heatmaps confirm that the model focuses on meaningful structures: ship hulls for true ships, clutter regions for false positives, and negligible areas for true negatives. Compared with prior binary-classification works, this study advances the field by addressing the three-class problem with rigorous validation and interpretability. This work has direct practical impact: by reducing false alarms and enabling fine-grained ship discrimination, it can enhance the reliability of automated maritime surveillance systems, aiding in illegal fishing detection, search and rescue operations, and maritime border security.

VI. LIMITATIONS AND FUTURE WORK

The dataset is limited to South African waters and Sentinel-1/RADARSAT-2 imagery; generalization to other regions and sensors remains untested. Patches are fixed at 51×51 pixels, which may not fully capture large vessels.

Future work will evaluate cross-sensor and cross-region performance, incorporate multi-scale inputs, and explore lightweight deployment for real-time surveillance. Integrating the model into a complete detection pipeline (CFAR pre-screening plus CNN discrimination) could further reduce false alarms.

VII. CONCLUSION

This paper introduced a three-class CNN for discriminating ships, false alarms, and clutter in SAR imagery. The model achieved 99.74% test accuracy and a macro F1 of 0.9967, with cross-validation confirming robustness ($99.63\% \pm 0.20\%$). Ablation studies and GradCAM visualizations validated the architectural choices and interpretability. This work offers a reliable, high-performance tool for maritime surveillance applications such as illegal fishing detection and border security.

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